# Somesh Khandelia, M.Sc.

Computer Vision · Machine Learning · Software Engineering
Munich, Germany · valid German work permit
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## EDUCATION

**M.Sc. Informatik (Computer Science)** April 2020 – April 2023
Technical University of Munich (TUM), Germany
Degree awarded April 2023, grade 1.7. Major: Computer Vision. Minors: Machine Learning, Software Engineering. Thesis grade 1.0.

**B.Tech. Computer Science & Engineering** August 2014 – May 2018
National Institute of Technology Durgapur, India
Degree awarded November 2018, grade 1.2 (German scale) / 9.52 of 10. Ranked 2nd in a department of 105.

**Indian School Certificate (Year 12, equivalent to Abitur)** 2011 – 2013
M. P. Birla Foundation Higher Secondary School, Kolkata, India
Awarded March 2013, result 95%.

**Indian Certificate of Secondary Education (Year 10)** 2000 – 2011
St. Aloysius Orphanage and Day School, Kolkata, India
Result 93.8%, ranked 1st in the school.

## PROFESSIONAL EXPERIENCE

**Wissenschaftlicher Mitarbeiter (Research Associate), 40 h/week** 01 May 2025 – 31 Oct 2025
TUM Klinikum rechts der Isar — Data Integration Center, Munich
Built a Python pseudonymisation library, CLI tool and FastAPI backend for clinical data. Developed a memory-efficient streaming parser converting >20 GB hierarchical DICOM JSON into FHIR, and a statistical analysis tool for large oncology XML exports.

**Software Developer, Amazon Robotics — Scanless Technology, full-time** October 2024 – December 2024
Amazon, Germany
Deployed a Python single-camera barcode scanner to production using Docker and AWS. Quantified production performance gaps on site in Nuremberg and Leipzig, holding automated scanning accuracy above 99%.

**Wissenschaftlicher Mitarbeiter (Research Associate), 40 h/week** 11 Mar 2024 – 31 May 2024
Technical University of Munich (TUM), Munich
Technical project lead for a three-person team building a t3-stack Next.js application supporting clinicians treating depression with patient-generated health data. IT services administration for a department of 10+; codebase owner on GitHub.

**Software Developer, 40 h/week** 01 Aug 2023 – 30 Sep 2023
Scandio GmbH, Munich
Client web development in TypeScript and React.js. Completed advanced coursework in Python, React.js, Redux and TypeScript, plus AWS Technical and Sales Accreditations.

**Wissenschaftler, Masterarbeit (Master's thesis researcher)** 17 Oct 2022 – 14 Apr 2023
Deutsches Zentrum für Luft- und Raumfahrt (DLR) — Earth Observation Center
Multi-vehicle detection and tracking in aerial image sequences using deep learning. Achieved the best overall MOT benchmark (HOTA, MOTA, IDF1) across both the DLR aerial and A9 ground datasets. Supervised by Prof. Dr. Alois Knoll, advised by Dr. Walter Zimmer. Grade 1.0.

**Werkstudent, Teilzeit (Working Student, Software Developer), 20 h/week** 01 Sep 2022 – 21 Apr 2023

TNG Technology Consulting GmbH, Munich

Frontend of OpossumUI, an open-source licence-compliance tool for the client Meta. Statistical visualisations in React.js, Redux, TypeScript and Recharts.

**Werkstudent, Teilzeit (Working Student, Machine Learning), 20 h/week** 26 Jan 2022 – 31 Jul 2022

Siemens AG — Digital Lab, Global Business Services, Munich

Wear detection in microscopic images of manufacturing tools via image registration and keypoint matching. Document search using OCR and PDF text mining. Job-description similarity analysis using BERT-based models.

**Praktikum (Internship), Web Developer** 06 Sep 2021 – 10 Dec 2021

Volkswagen Data Lab, Munich

Frontend of a battery-testing web application in React.js, TypeScript and GraphQL, covering supplier, user and data management.

**Werkstudent, Teilzeit (Working Student, Machine Learning), 20 h/week** 07 Jun 2021 – 06 Dec 2021

Rohde & Schwarz, Munich

Machine learning models predicting optimal cabling scenarios between a test system and multiple devices under test, replacing manual work with complex user manuals.

**Werkstudent, Teilzeit (Working Student, Data Engineer), 20 h/week** 01 Sep 2020 – 31 May 2021

Fusionbase, Munich

More than 15 data pipelines using web scraping and multiprocessing in Python, feeding structured datasets into MySQL and MongoDB. Reduced a 52-million-row GDELT update from 8 hours to 25 minutes. Clients included BMW.

**Wissenschaftlicher Mitarbeiter (Research Associate), 40 h/week** 26 Aug 2019 – 22 Nov 2019

Indian Institute of Technology Kharagpur — CNeRG, India

Intel-sponsored research using deep learning to predict cellular network throughput, reducing battery consumption for video streaming by approximately 30%. Supervised by Prof. Dr. Sandip Chakraborty.

**Software Developer, Options Trading, 40 h/week** 04 Jun 2018 – 28 Jun 2019

D. E. Shaw & Co., Hyderabad, India

Revamped the infrastructure for the firm's US options strategy covering 900,000 instruments, using OOP and test-driven development with over 90% code coverage. Two-week DevOps rotations across global exchanges and corporate actions.

## PUBLICATIONS

**Multi-Vehicle Detection and Tracking in Aerial Image Sequences based on Deep Learning** 2023

Master's Thesis, TUM & DLR · elib.dlr.de/194656

**EnDASH — A Mobility Adapted Energy Efficient ABR Video Streaming for Cellular Networks** 2020

IFIP Networking Conference · ieeexplore.ieee.org/document/9142696

## SKILLS

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|---|---|
| **Programming** | Python, C++, TypeScript, C# |
| **AI & data** | PyTorch, TensorFlow, HuggingFace, scikit-learn, OpenCV, NumPy, pandas, SciPy |
| **Web** | React.js, Next.js, Redux, FastAPI, Node.js, Express, REST, GraphQL |
| **Data stores** | PostgreSQL, MySQL, MongoDB |
| **DevOps** | Git, Docker, AWS, Scrum, test-driven development |
| **Languages** | English (C2), German (B2), Hindi (mother tongue) |

## AWARDS

Deutschlandstipendium (German National Scholarship) 2020–2021 · e-fellows.net scholarship · hackaTUM 2020 finalist · 2nd runner-up, firmwide hackathon at D. E. Shaw